

A family health record platform for children with special needs, designed to centralize medications, appointments, therapy goals, care team information, and children's health records.

HEALTH WELLNESS

PRODUCT DESIGN CASE STUDY

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A CENTRAL PLACE FOR HEALTH MANAGEMENT

PRODUCT DESIGN CASE STUDY

A family health record platform for children with special needs, designed to centralize medications, appointments, therapy goals, care team information, and children's health records.

Product Design | UX/UI | Interaction Design | Prototyping | Implementation

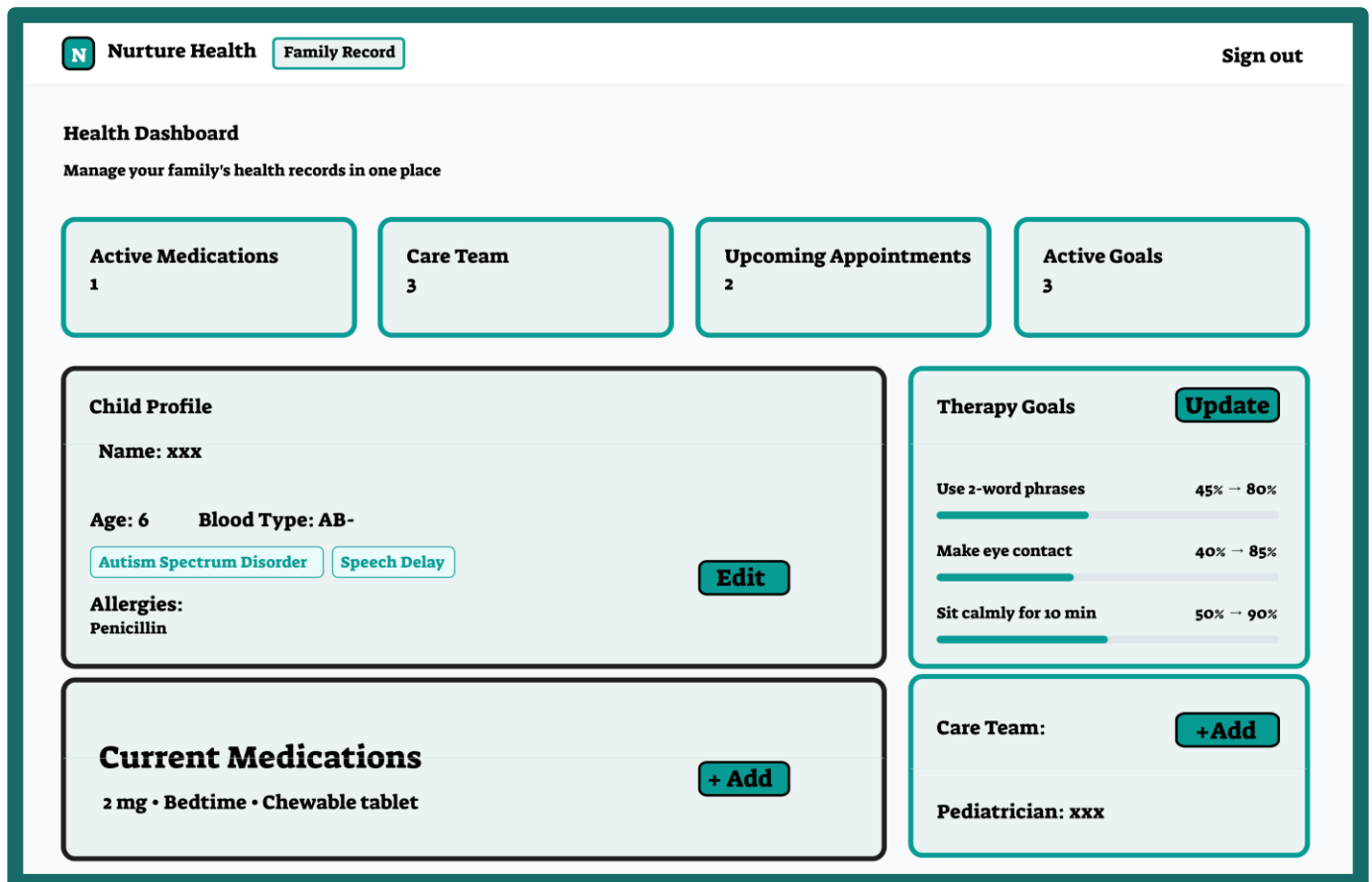
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01 — PRODUCT

THE PRODUCT

Health-Wellness is a complete family health record platform for children with special needs. The product provides a centralized dashboard that lets parents and caregivers manage medications, appointments, therapy goals, care team members, and the child's core health information.



The image is a UI mockup of the Health-Wellness dashboard. At the top, there is a header bar with the 'Nurture Health' logo on the left, a 'Family Record' tab in the center, and a 'Sign out' link on the right. Below the header, the main content area is titled 'Health Dashboard' with the subtitle 'Manage your family's health records in one place'. This area is divided into several sections. At the top, there are four summary cards: 'Active Medications' with a count of 1, 'Care Team' with a count of 3, 'Upcoming Appointments' with a count of 2, and 'Active Goals' with a count of 3. Below these, the dashboard is split into two columns. The left column contains a 'Child Profile' section with fields for Name (xxx), Age (6), Blood Type (AB-), and two tags for 'Autism Spectrum Disorder' and 'Speech Delay'. It also lists an allergy to 'Penicillin' and includes an 'Edit' button. Below the profile is a 'Current Medications' section showing '2 mg • Bedtime • Chewable tablet' and an '+ Add' button. The right column features a 'Therapy Goals' section with an 'Update' button and three progress bars for 'Use 2-word phrases' (45% to 80%), 'Make eye contact' (40% to 85%), and 'Sit calmly for 10 min' (50% to 90%). At the bottom of the right column is a 'Care Team' section with a '+ Add' button and the text 'Pediatrician: xxx'.

N Nurture Health

Family Record

Sign out

Health Dashboard

Manage your family's health records in one place

Active Medications
1

Care Team
3

Upcoming Appointments
2

Active Goals
3

Child Profile

Name: xxx

Age: 6 Blood Type: AB-

Autism Spectrum Disorder Speech Delay

Allergies:
Penicillin

Edit

Current Medications

2 mg • Bedtime • Chewable tablet

+ Add

Therapy Goals Update

Use 2-word phrases

45% → 80%

Make eye contact

40% → 85%

Sit calmly for 10 min

50% → 90%

Care Team:

+ Add

Pediatrician: xxx

Health-Wellness Dashboard - Centralized view of medications, appointments, therapy goals, and care team information.

02 — CHALLENGES

Design Challenge

The product addresses a coordination problem: families may need to keep track of multiple providers, medications, appointments, therapy goals, and health-record details. The design opportunity is to reduce fragmentation and make the most important information visible briefly.

USER PROBLEM

- Health information is scattered across sources
- Parents need to coordinate several care activities
- Progress is difficult to understand without visual feedback
- Updating information should be quick and low friction

PRODUCT OPPORTUNITY

- Centralized dashboard
- Clear information hierarchy
- Simple forms and feedback
- Visual progress tracking
- Care-team directory

Product Scope

- Secure authentication
- Health dashboard
- Medication tracking
- Appointment scheduling
- Therapy goal progress
- Care-team directory
- Child profile management
- Centralized dashboard

03 — RESEARCH

RESEARCH EXECUTION

The project documentation reports that interviews and surveys were used to understand how parents manage their child's healthcare information. The research identified information overload, scattered records, missed appointments, and the need for a single place to manage health information as recurring themes.

Research finding	Reported result
Parents feeling stressed managing medical/therapy information	85%
Parents who had missed at least one appointment	70%
Parents wanting a single place for medications, appointments, and progress	90%
Parents who would use an app to simplify healthcare management	60%

What users wanted

- A simple dashboard showing everything immediately
- Visual progress tracking using charts or progress bars
- Easy medication and appointment management
- A central place to store information
- Simple language instead of confusing medical jargon
- Shareable reports for doctors

04 — USER INTERVIEWS

INTERVIEWS

Sarah - 34

Question	Response
Occupation	Marketing Manager
Current tools	Phone calendar, notebook, spreadsheets
Obstacles	Too many appointments; scattered information; no central progress tracking
Desired features	Dashboard, medication tracking, reminders, progress charts, provider information
Visual progress	Would help determine whether therapy is working

Michael - 41

Question	Response
Occupation	High School Teacher
Current tools	Paper notes and spreadsheet
Obstacles	Too much information; confusing medical terms; no easy way to see progress
Desired features	Simple language, readable charts, clear indicators, notes, shareable reports
Visual progress	Would help understand whether therapy is working

04 — USER INTERVIEWS

INTERVIEW SYNTHESIS

The interviews consistently pointed to a common pattern: parents are overwhelmed by the volume of healthcare information they need to manage across multiple providers, and they struggle to track progress in a meaningful way. This pattern helped define the core design priorities: simplifying the experience, reducing friction for regular updates, and making progress visible at a glance.

RECURRING PAIN POINTS

- Information overload
- Fragmented tracking
- Appointment management
- Unclear progress

DESIGN IMPLICATIONS

- Prioritize overview
- Use plain language
- Make updates lightweight
- Make progress visible
- Support provider information

05 — SURVEYS

SURVEY INSIGHTS

The project documentation reports survey responses about provider counting, medication tracking, appointment frequency, and what would make healthcare management easier.

Question	Response distribution
How many healthcare providers?	1: 10% 2–3: 30% 4–5: 40% 6+: 20%
How are medications tracked?	Paper: 45% Phone reminders: 30% Spreadsheet: 15% Nothing: 10%
How often are appointments missed?	Never: 20% Rarely: 40% Sometimes: 30% Often: 10%
What would make management easier?	Central place: 60% Visual progress: 20% Reminders: 15% Sharing: 5%

Key Opportunity

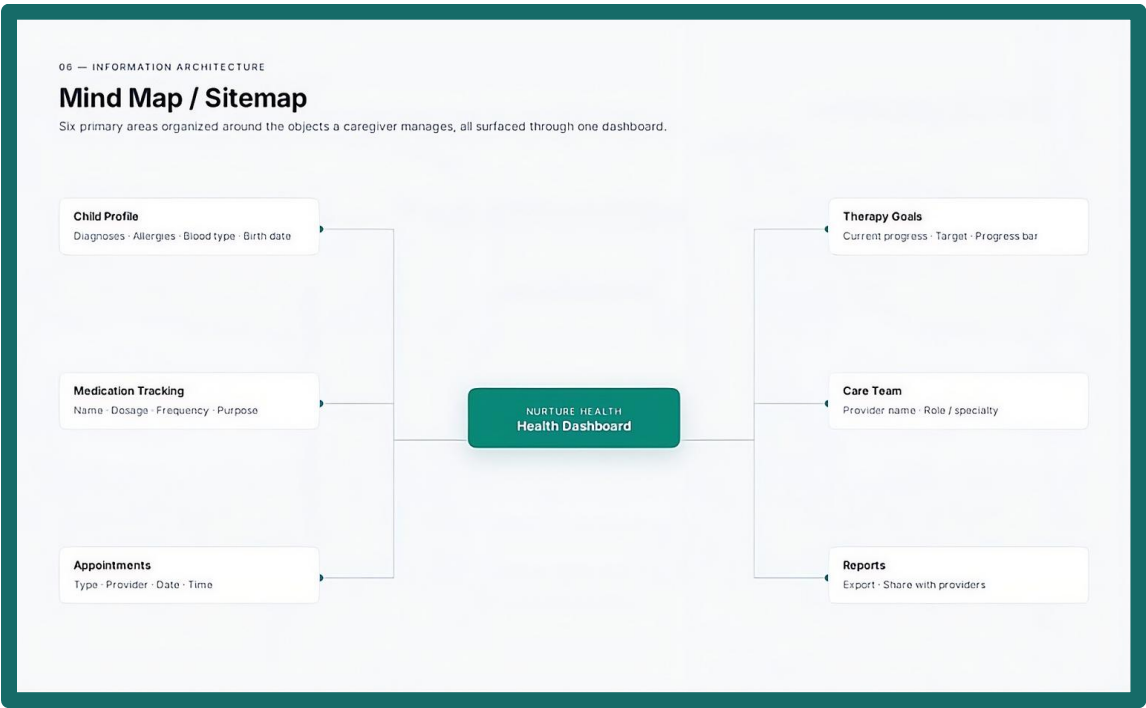
The strongest preference reported was a single central place for healthcare information. This supports the decision to make the dashboard the product's primary organizing surface.

06 — ARCHITECTURE

INFORMATION ARCHITECTURE

The information architecture organizes the product into five primary areas: child profile, medications, appointments, therapy goals, and care team, all accessible through a single dashboard.

Primary area	Key information
Child Profile	Diagnoses, allergies, blood type
Medication Tracking	Dosage, frequency, purpose
Appointments	Date, provider, notes
Therapy Goals	Progress, target
Care Team	Providers, specialties
Reports	Export, share



Information Architecture / Mind Map

07 — USER FLOWS

PRIMARY USER FLOWS

Flow A — Add a Medication

Login → Dashboard → Add Medication → Enter medication name, dosage, frequency, and purpose → Save → Medication appears in the list.

Flow B — Update Therapy Goal

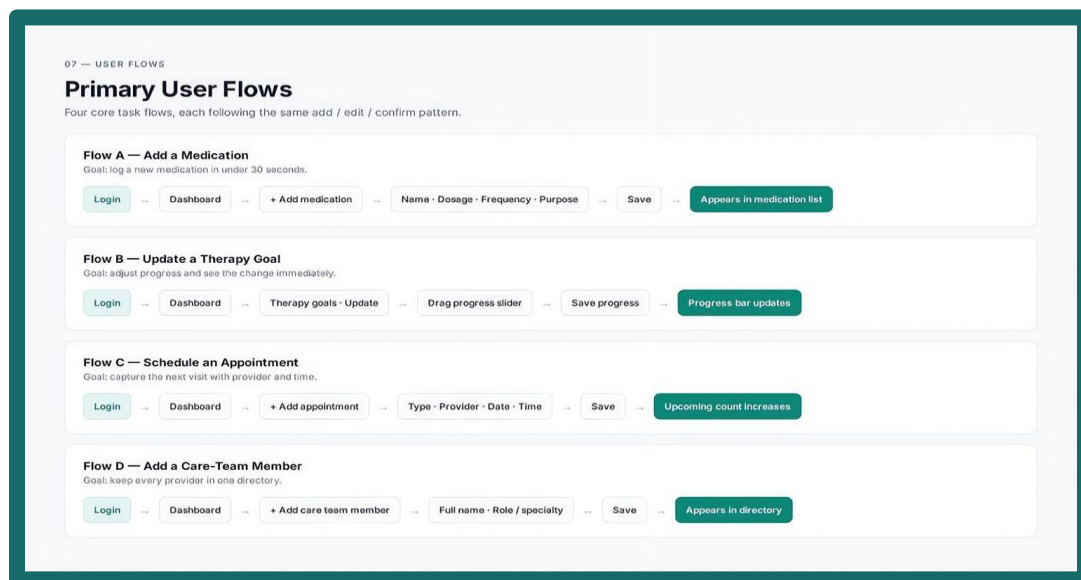
Login → Dashboard → Therapy Goals → Update → Adjust progress → Save Progress → Progress indicator updates.

Flow C — Schedule an Appointment

Login → Dashboard → Schedule Appointment → Enter appointment type, provider, date, and time → Save → Appointment appears in the schedule.

Flow D — Add a Care-Team Member

Login → Dashboard → Add Care Team Member → Enter provider details → Save → Provider appears in the directory.



User Flow Diagram

08 — PERSONAS

PERSONA 1 - SARAH JOHNSON

Attribute	Profile
Age	34
Role	Primary caregiver/parent
Profession	Marketing Manager
Child	Alex, age 6
Characteristics	Comfortable with technology; uses smartphone and laptop daily; organized and detail-oriented
Pain points	Multiple providers; scattered information; missed appointments; unclear therapy progress
Needs	Centralized health information; visual progress; reminders; simple experience

Daily Routine

7:00 AM — Prepares breakfast.

8:00 AM — Takes Alex to school.

9:00 AM — Works.

12:00 PM — Checks phone for therapy updates.

3:00 PM — Picks Alex up.

4:00 PM — Opens Health-Wellness, adds medication and checks appointments.

8:00 PM — Updates therapy goal progress.

08 — PERSONAS

PERSONA 2 - MICHAEL CHEN

Attribute	Profile
Age	41
Role	Secondary caregiver/parent
Profession	High School Teacher
Child	Emily, age 8
Characteristics	Comfortable with technology; prefers simple, clean interfaces; patient and detail-oriented
Pain points	Confusing medical terms; no easy progress view; information scattered
Needs	Plain language; readable charts; clear indicators; shareable reports

Daily Routine

7:30 AM — Prepares breakfast.

8:30 AM — Takes Emily to school.

9:00 AM — Works.

3:30 PM — Picks Emily up.

4:00 PM — Checks therapy progress.

7:00 PM — Updates therapy goals.

9:00 PM — Winds down.

09 — PRODUCT DEFINITION

USER STORIES

- As a parent, I want a simple dashboard to see my child's health information briefly.
- As a parent, I want to track medications and get reminders.
- As a parent, I want to see therapy progress with visual charts.
- As a parent, I want to store information provided in one place.
- As a parent, I want to share reports with doctors easily.
- As a parent, I want to add and update information quickly.

JOURNEY MAP

Stage	Thinking	Doing	Experience goal
Login	I want secure access.	Enter credentials.	Welcoming, secure
Dashboard	I want to see everything at a glance.	Review overview.	Organized, clear
Child Profile	I want to check health information.	Open profile.	Informative, complete
Medication	I need to add a medication.	Enter details.	Simple, fast
Therapy Goal	I want to track progress.	Adjust progress.	Visual, motivating
Appointments	I need upcoming visits.	Open schedule.	Clear, organized

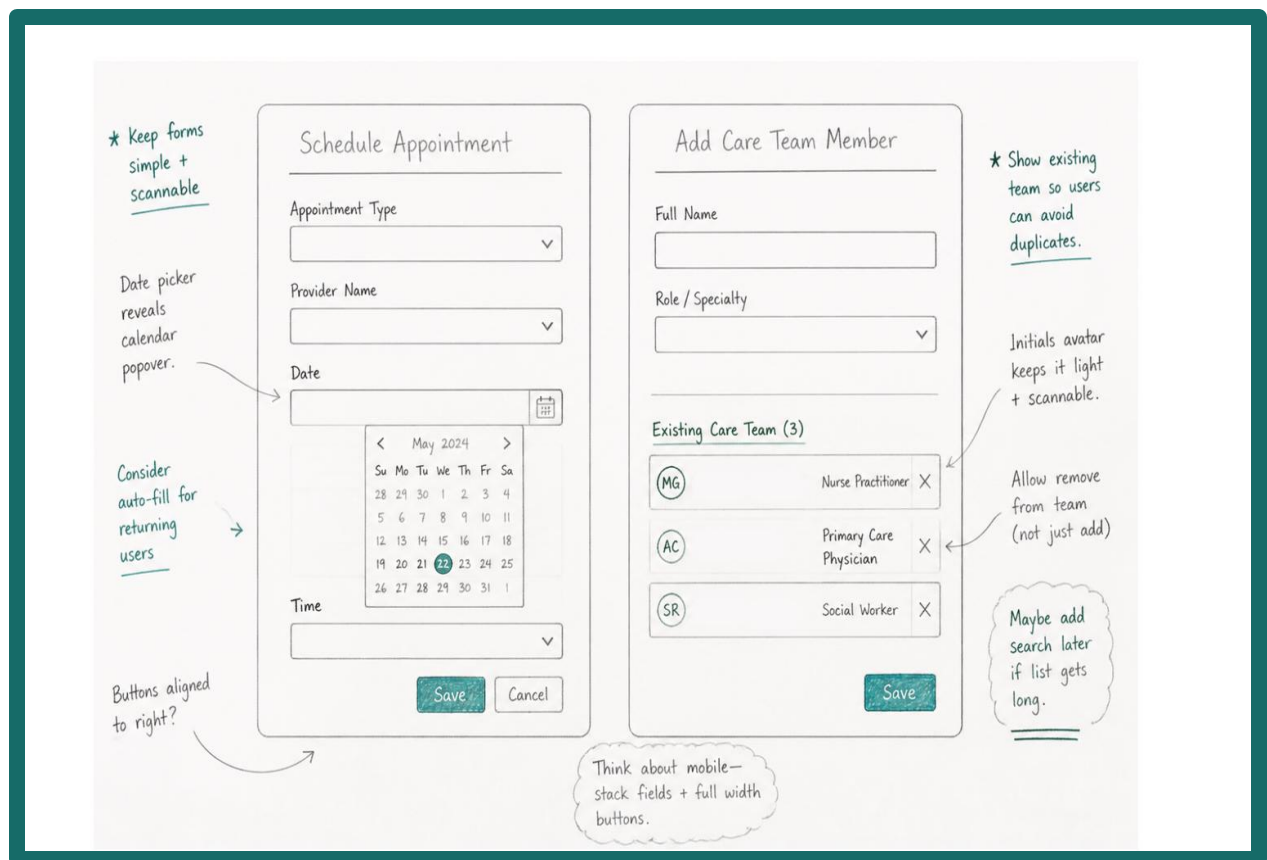
10 — SKETCH

DESIGN EXPLORATION SKETCH

Early design notes for the Schedule Appointment and Add Care Team forms, capturing decisions later reflected in Iterations #4 and #5.

Before moving into high-fidelity wireframes, exploratory sketches were created to test layout options, field grouping, and interaction patterns for two key tasks: scheduling an appointment and adding a care team member.

These sketches were intentionally to evaluate structure and flow before investing in visual polish.



Product - Sketch

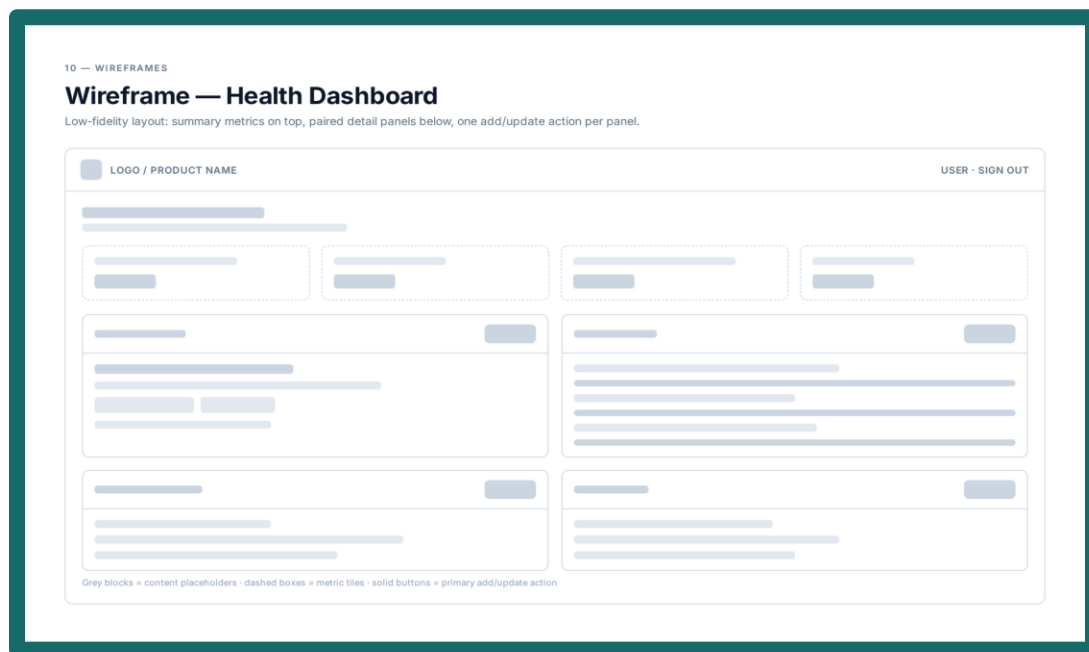
11 — WIREFRAMES

WIREFRAMES

The wireframing phase focuses on the core interaction flows rather than visual polish. The main question is whether caregivers can complete important health-management tasks with a logical sequence and minimal friction.

Health - Dashboard Wireframe

This low-fidelity wireframe establishes the foundational layout of the health-Wellness dashboard. The design prioritizes clear information hierarchy: key metrics (medication count, care team size, upcoming appointments, active goals) are placed at the top, followed by detail panels for therapy goals and recent activity. The goal is to give caregivers an immediate, at-a-glance understanding of their child's health status without unnecessary navigation.

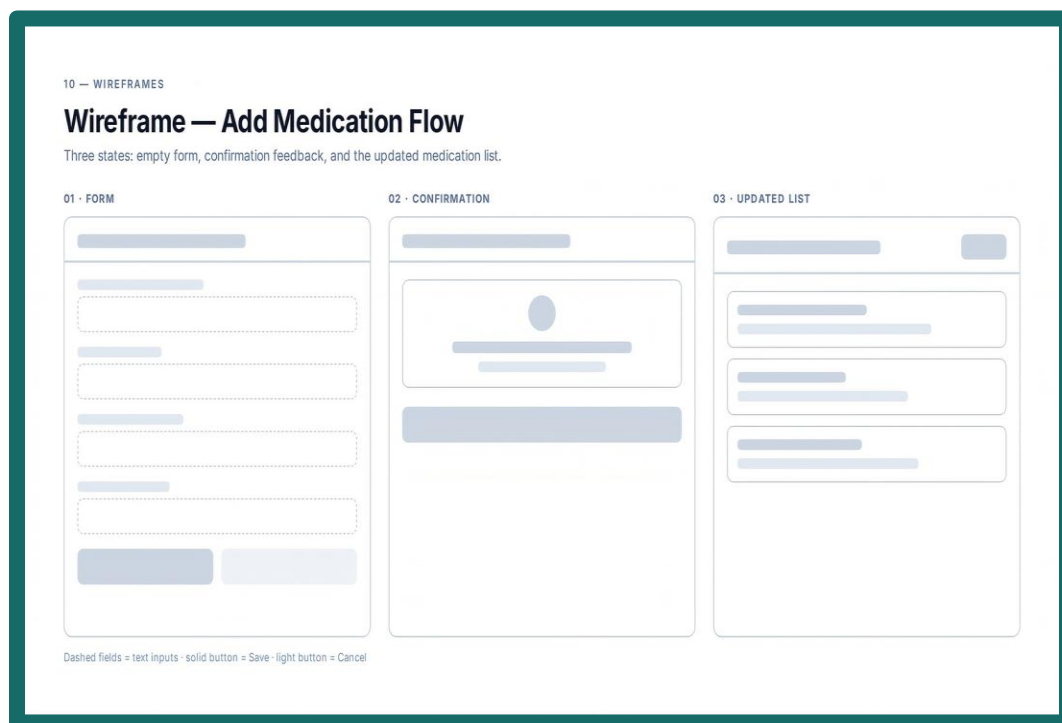


Wireframe - Dashboard

11 — WIREFRAMES

Health - Add Medication Wireframe

This wireframe maps the step-by-step flow for adding new medication. The form is structured to collect essential information- name, dosage, frequency, and purpose- in a logical order. The design emphasizes clarity and speed, allowing caregivers to complete the task in under 30 seconds. A confirmation state is included to reassure users that the medication has been successfully saved.

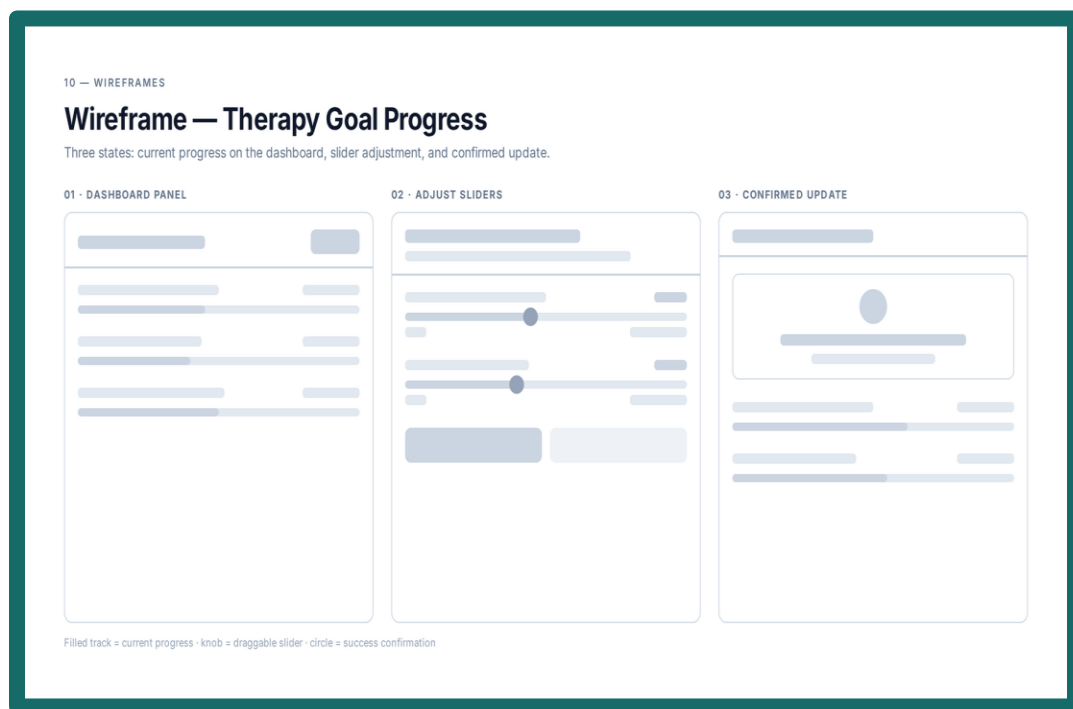


Wireframe – Add Medication

11 — WIREFRAMES

Health - Therapy Progress Wireframe

This wireframe illustrates the process of updating a therapy goal. The design uses a visual slider to adjust progress percentages, providing immediate visual feedback as the user interacts. The goal is to make tracking progress feel tangible and motivating. A clear target is displayed alongside the current level, helping caregivers understand how close their child is to achieving each goal.



Wireframe - Therapy Progress

12 — ITERATION #1

ITERATION #1 - LOGIN & ONBOARDING

Users can securely access their account and understand the product's purpose immediately upon entry.

What Was Tested

The login and account-access process.

Key Design Decisions

Decision	Rationale
• Simple login form • Clear demo credentials • Branded, calm visual • Validation feedback	• Only email and password fields to reduce cognitive load • Allows reviewers to test the product without creating an account • Uses the Nurture Health identity to build trust • Clear error messages for invalid credentials

12 — ITERATION #1



The mockup shows a login screen for 'Nurture Health'. At the top center is a small teal square icon with a white 'N'. Below it, the text 'Nurture Health' is centered in a bold, black, sans-serif font. Underneath that, the tagline 'Complete family health record' is centered in a smaller, regular, black, sans-serif font. The login form consists of three main elements: a 'Username' label followed by a text input field containing 'parent@example.com'; a 'Password' label followed by a password input field with ten dots; and a large teal button with the text 'Sign in' in white. Below the button is a horizontal line, and underneath that, the text 'Demo credentials' and 'parent@example.com' are centered.

Iteration #1 - Login screen

Onboarding Flow

- User enters email and password.
- System validates credentials.
- Upon success, the user is directed to the dashboard.
- Invalid credentials trigger a clear, helpful error message.

What Users Liked

- Simple, uncluttered interface
- Demo credentials made testing easy
- Clear visual hierarchy

What Was Improved

- Added validation feedback
- Simplified the form further
- Reinforced the brand identity

12 — ITERATION #2

ITERATION #2 - ADD MEDICATION

What was tested

The process of adding medication.

Goal

Users can add medication with dosage, frequency, and purpose.

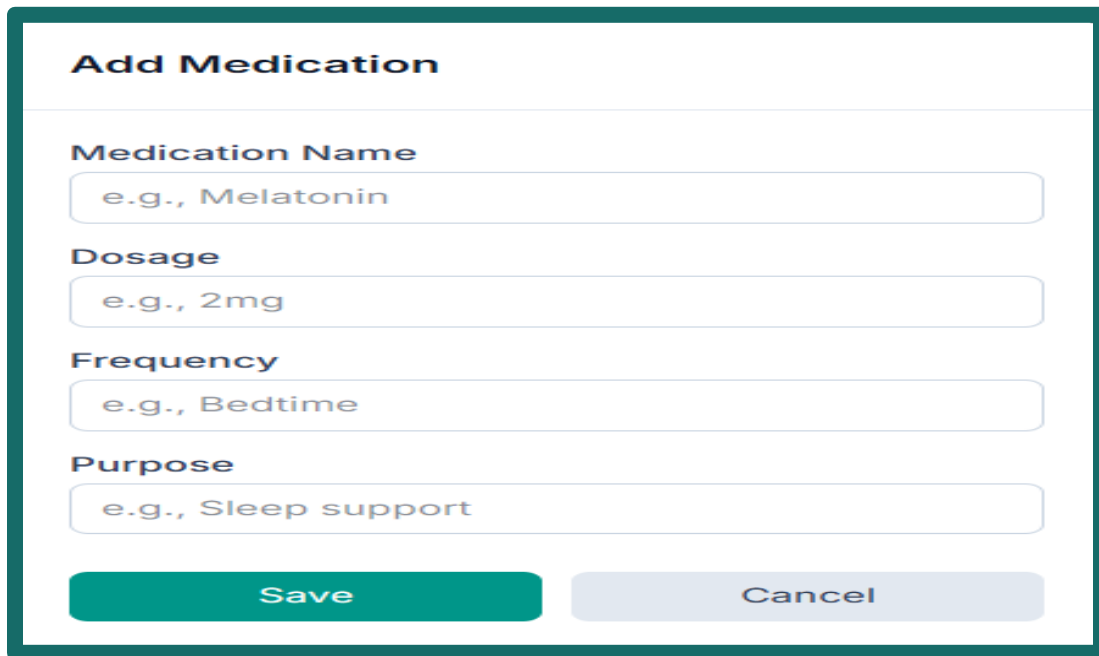
Test script

Login → Add Medication → Fill in details → Save.

Results

- The overall progression made sense.
- Some steps were confusing.
- A confirmation message was needed.

Simplify the form and add clearer labels and confirmation feedback.



Add Medication

Medication Name
e.g., Melatonin

Dosage
e.g., 2mg

Frequency
e.g., Bedtime

Purpose
e.g., Sleep support

Save **Cancel**

Iteration #2 – Add Medication

12 — ITERATION #3

ITERATION #3 - MEDICATION FLOW

What was tested

The improved medication-entry process.

Results

- The flow made sense.
- The simplified form removed unnecessary steps.
- A confirmation modal was added.

Design response

Keep the reduced form complexity and reinforce completion with clear confirmation.

Add Medication

✓ Medication saved successfully

Medication Name
Melatonin

Dosage
2mg

Frequency
Bedtime

Purpose
Sleep support

Save **Cancel**

Iteration #2 — Medication flow with confirmation feedback

Iteration #3 - Medication Flow

12 — ITERATION #4

ITERATION #4 - THERAPY PROGRESS

What was tested

The process of updating a therapy goal.

Goal

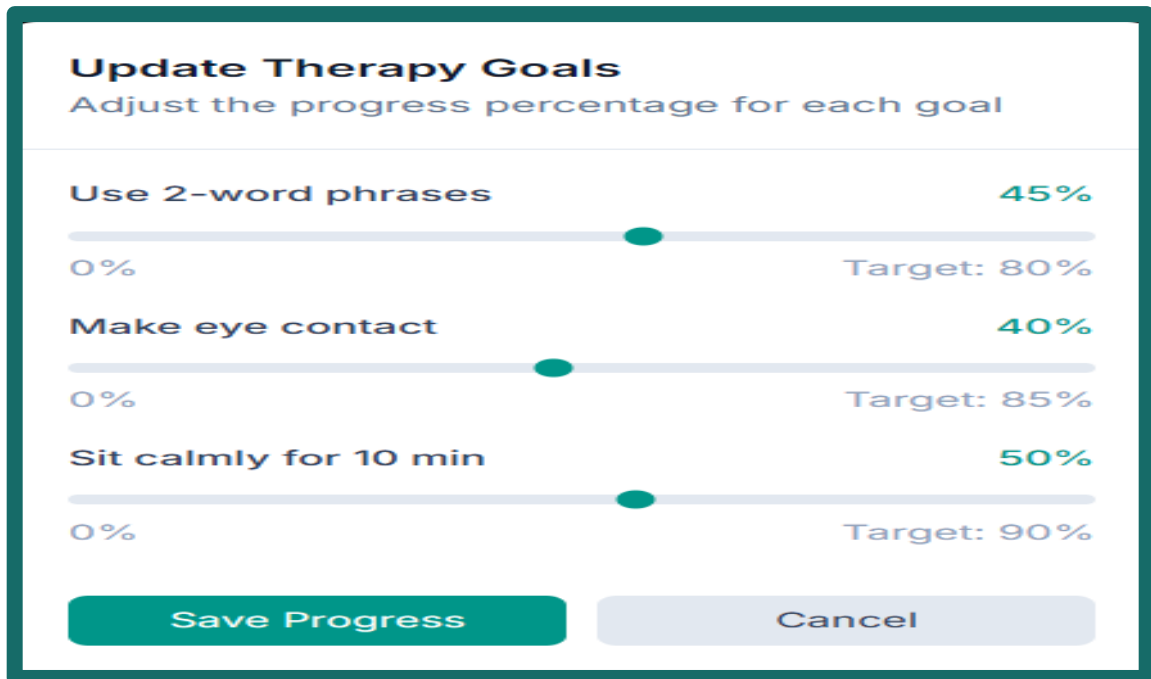
Users can update therapy goal progress.

Results

- The sequence made sense.
- No unnecessary steps were identified.
- Users needed stronger visual feedback.

Design response

Add a visual progress bar and confirmation message so users can immediately understand the result of an update.



The screenshot shows a mobile app interface titled "Update Therapy Goals" with the subtitle "Adjust the progress percentage for each goal". It features three horizontal progress bars, each with a green slider dot. The first goal is "Use 2-word phrases" at 45% progress, with a target of 80%. The second is "Make eye contact" at 40% progress, with a target of 85%. The third is "Sit calmly for 10 min" at 50% progress, with a target of 90%. At the bottom are two buttons: "Save Progress" (green) and "Cancel" (grey).

Goal	Current Progress	Target
Use 2-word phrases	45%	80%
Make eye contact	40%	85%
Sit calmly for 10 min	50%	90%

Iteration #4 - Therapy Progress

12 — ITERATION #5

ITERATION #5 - APPOINTMENT SCHEDULING

What was tested

The process of scheduling an appointment with a provider.

Results

- The flow was logical.
- No unnecessary steps were identified.
- A date picker was needed for easier scheduling.

Design response

Introduce a clearer date-selection interaction.

Schedule Appointment

Appointment Type
e.g., Pediatrician, Dentist, Therapy

Provider Name
e.g., Dr. Smith

Date
e.g., April 10, 2025

Time
e.g., 3:00 PM

Save **Cancel**

Iteration #5 – Appointment Scheduling

12 — ITERATION #6

ITERATION #6 - CARE TEAM

What was tested

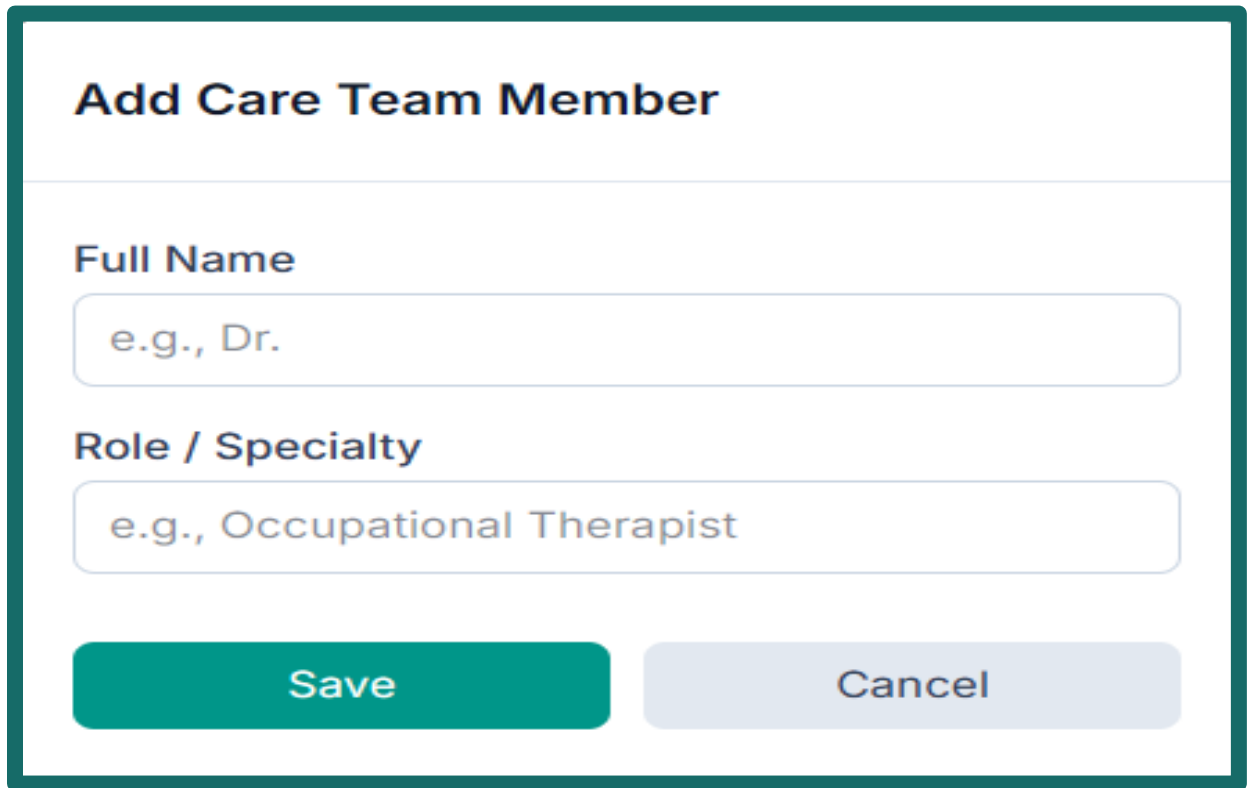
The process of adding a care team member.

Results

- The flow made sense.
- The progression was logical.
- No additional changes were identified.

Design response

Finalize the care-team flow and maintain consistency with the other add/edit patterns.



Add Care Team Member

Full Name

e.g., Dr.

Role / Specialty

e.g., Occupational Therapist

Save **Cancel**

Iteration #6 – Care Team

13 — SUMMARY

ITERATION SUMMARY OVERVIEW

Across five iterations, the design was refined through usability testing and user feedback. Key improvements included simplifying the medication form, adding confirmation feedback, introducing a date picker for appointments, and establishing consistent add/edit patterns. The image below summarizes the final interaction flows that were carried into the high-fidelity prototype.

11 — ITERATIONS

Iteration Summary (1-5)

What was tested, what testing revealed, and how the design responded across five rounds.

#	FOCUS	RESULT FROM TESTING	DESIGN RESPONSE	STATUS
01	Add medication	Form felt long; labels were unclear and users were unsure the entry saved.	Simplify the form, clarify labels, add confirmation feedback.	Revised
02	Medication flow	Reduced form completed faster; users still wanted explicit completion.	Keep the reduced complexity, add a confirmation modal.	Revised
03	Therapy goal	Users could update progress but could not see the effect of the change.	Add a visual progress bar and an immediate confirmation message.	Revised
04	Appointment scheduling	Typing dates by hand caused hesitation and formatting errors.	Introduce a clearer date-selection interaction (date picker).	Revised
05	Care team	Flow completed without difficulty; no additional changes identified.	Finalize the flow and keep it consistent with other add/edit patterns.	Validated

Iteration Summary

FINAL PRODUCT EXPERIENCE

The final implemented product brings the validated interaction patterns into a functional interface. The live project currently presents the following product areas.

Feature	Experience
Secure Authentication	Login with validation and clear invalid-credential feedback.
Health Dashboard	Overview of active medications, care-team size, upcoming appointments, and active therapy goals.
Medication Tracking	Add and manage medication name, dosage, frequency, and purpose.
Appointment Scheduler	Store appointment type, provider, date, and time.
Therapy Goal Progress	Update goal progress using interactive sliders and visual progress bars.
Care Team Directory	Add doctors, therapists, and specialists with names and roles.
Child Profile	Manage name, birth date, blood type, allergies, and diagnoses.

15 — PRODUCT THINKING

KEY DESIGN DECISIONS

01 — DASHBOARD FIRST

- Make the product's highest-value information visible immediately
- Reduce navigation for common monitoring tasks

02 — STRUCTURED FORMS

- Keep health-data entry explicit and consistent
- Use labels that map directly to user needs

03 — VISUAL PROGRESS

- Turn therapy updates into an immediately readable status
- Use progress bars to support quick interpretation

04 — CONSISTENT PATTERNS

- Reuse add/edit/save interaction patterns
- Reduce the learning curve across features

Product Value Proposition

Health-Wellness turns fragmented family healthcare-management tasks into a single, structured product experience. The value lies not only in the individual features; it lies in the connections among them through a single dashboard and a consistent information model.

USABILITY & ACCESSIBILITY CONSIDERATIONS

The product direction supports a healthcare context where clarity and error prevention are especially important. The following considerations should guide future refinement.

- Use plain, unambiguous labels for health data.
- Provide clear validation and error feedback.
- Make primary actions visually distinct.
- Maintain readable text and sufficient contrast.
- Avoid relying on colour alone to communicate progress or status.
- Confirm important data changes.
- Keep add/edit interactions predictable across modules.
- Design responsive layouts for desktop and mobile use.

Design System Direction

Area	Direction
Color	Calm healthcare palette with teal as an accent
Typography	High readability and clear hierarchy
Components	Reusable cards, forms, modals, buttons, progress indicators
Layout	Dashboard-oriented, scannable information blocks
Interaction	Immediate feedback after creating/update actions

FROM PRODUCT DESIGN TO IMPLEMENTATION

A key strength of this project is that the product design is connected to a working implementation rather than remaining only a static visual concept. The live portfolio describes Health-Wellness as a complete family health record platform and documents its functional flows.

Technology	Role in implementation
Next.js 16	Application framework
TypeScript	Typed application development
Tailwind CSS	UI styling
shadcn/ui	Reusable interface components
NextAuth.js	Authentication
Prisma	Database access layer
SQLite	Data storage
Lucide React	Interface icons

Design-Engineering Connection

The product's forms, dashboard statistics, progress updates, and editable records are implemented as functional interactions. This makes the project well-suited to demonstrate both product design thinking and front-end execution.

CONCLUSION

Health-Wellness explores how product design can simplify a complex coordination problem for families managing special-needs healthcare. The project moves from documented research findings to user flows, personas, journey mapping, wireframing, iterative refinement, and a functional final product.

What this project demonstrates

- Problem framing around a real user need
- Research-informed feature definition
- Information architecture and user flows
- Persona and journey-map development
- Iterative interaction design
- UI design and reusable patterns
- Functional front-end implementation
- Ability to connect product thinking with engineering execution

Next Steps

- Conduct additional usability testing with a broader caregiver sample.
- Validate the dashboard information hierarchy with real users.
- Explore push notifications for appointments and medications.
- Add exportable reports for healthcare providers.
- Explore more advanced progress visualization.
- Evaluate accessibility with formal WCAG-oriented testing.

FINAL PRODUCT DESIGN

Final Product Experience

Selected high-fidelity screens from the implemented Health-Wellness platform

Health Dashboard

N Nurture Health

Family Record

Sign out

Health Dashboard

Manage your family's health records in one place

Active Medications
1

Care Team
3

Upcoming Appointments
2

Active Goals
3

Child Profile

Age: • Blood Type:

Autism Spectrum Disorder Speech Delay

Allergies:
Penicillin

Edit

Therapy Goals

Update

Use 2-word phrases 45% → 80%

Make eye contact 40% → 85%

Sit calmly for 10 min 50% → 90%

Current Medications

+ Add

2 mg • Bedtime • Chewable tablet

Care Team

+ Add

Pediatrician

Sign In

N

Nurture Health

Complete family health record

Email address

Password

Sign in

Demo credentials
parent@example.com / password123

Edit Child Profile

Edit Child Profile

Child's Full Name

Birth Date

Blood Type

Allergies (comma separated)

Add New Diagnosis (optional)

Current Diagnoses:
Autism Spectrum Disorder Speech Delay

Save Changes Cancel

Final Prototype / Case Study Hero

APPENDIX

PROJECT REFERENCE

This product design case study was developed as an independent exploration of how healthcare coordination tools can better serve families of children with special needs. The concept originated from work at UGTS, and the goal was to reduce the fragmentation of healthcare information for caregivers.

Health-Wellness addresses this through a centralized dashboard that tracks medications, appointments, therapy goals, and care-team members.

Live product and implementation details should be kept synchronized with the portfolio version when this case study is published.